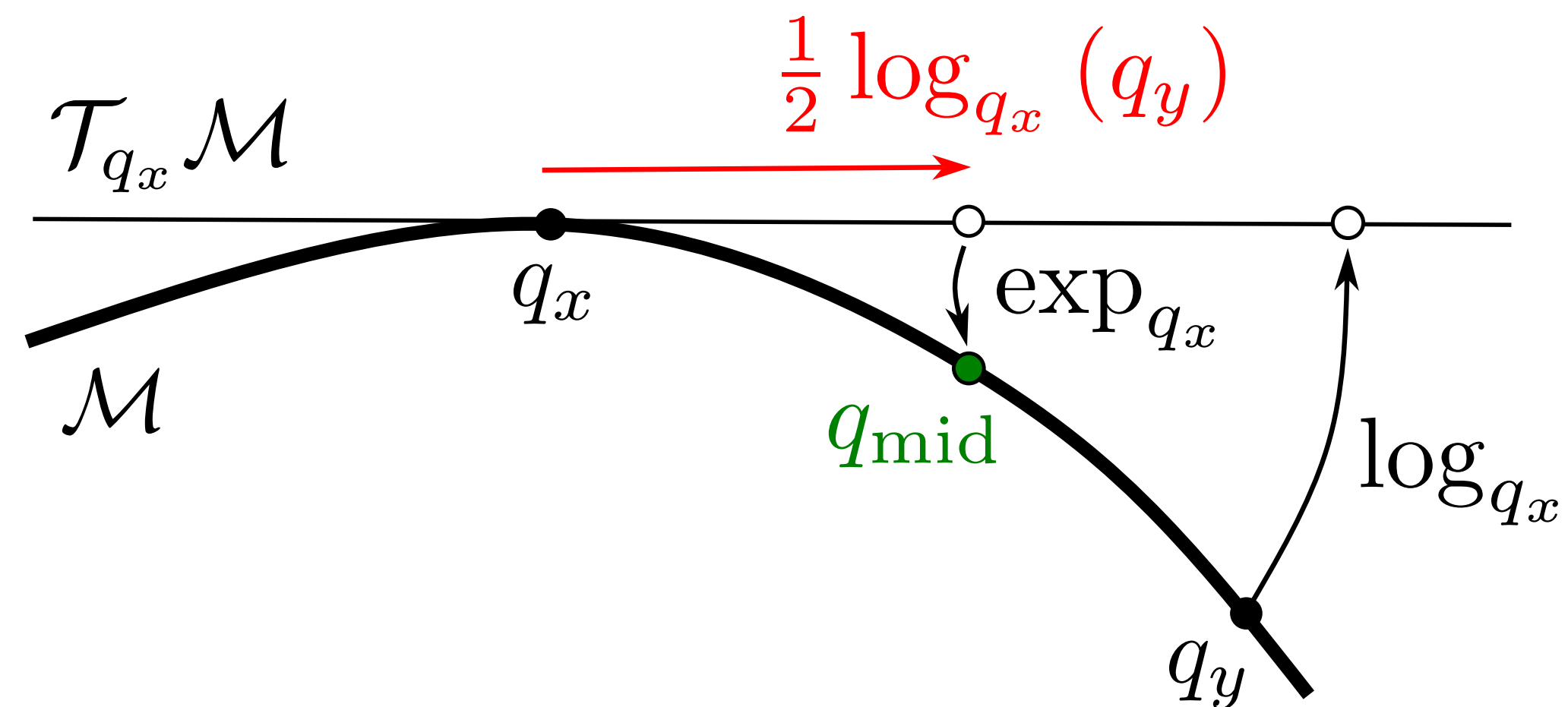


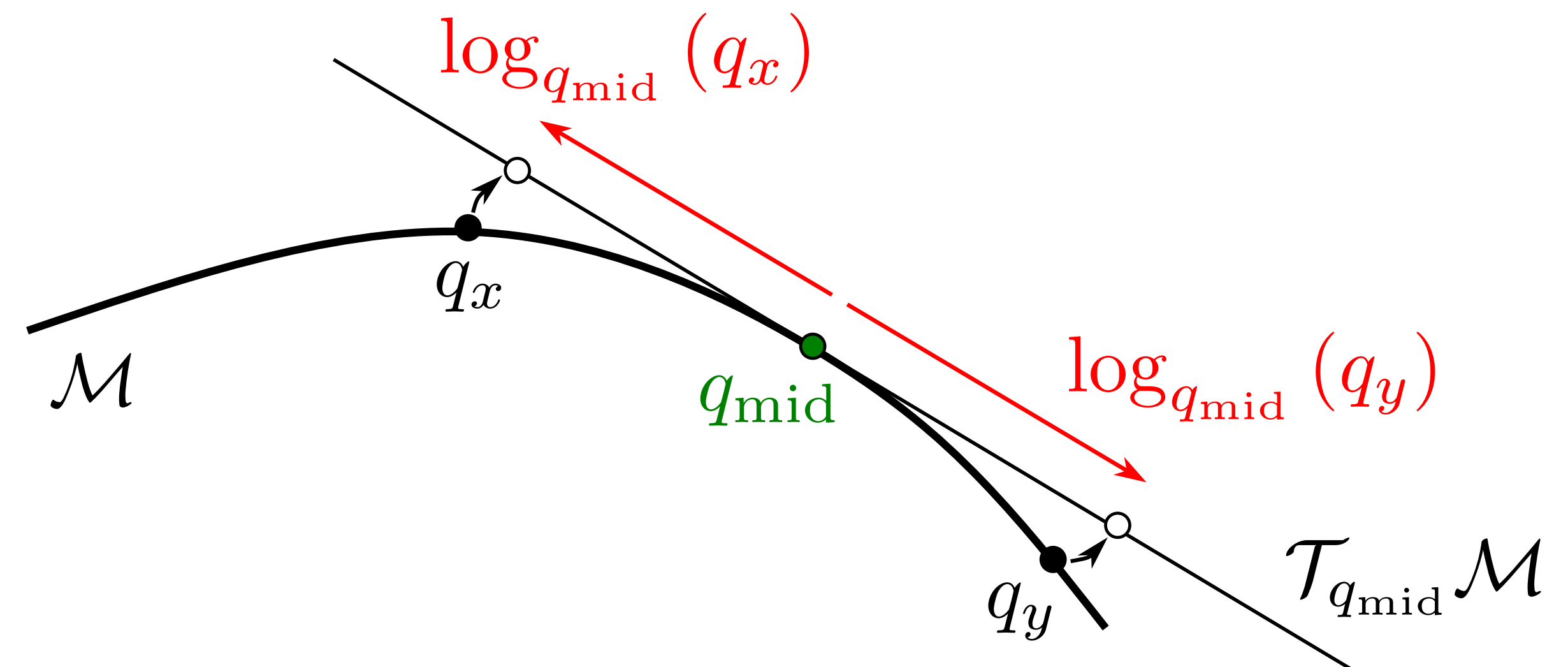
Midpoint configuration

$$q_{\text{mid}} = \exp_{q_x} \left(\frac{1}{2} \log_{q_x} (q_y) \right)$$



Riemannian distance — exact

$$d_{\mathcal{M}}(q_x, q_y) = \left\| \log_{q_{\text{mid}}}(q_y) - \log_{q_{\text{mid}}}(q_x) \right\|_{q_{\text{mid}}}$$



Replace the exp / log maps with **retractions** $\mathcal{R}_q(v), \mathcal{R}_q^{-1}(v)$

first-order • **cheap** to compute

Discrete geodesic interpolation

— iteratively descends along the negative **Riemannian natural-gradient** direction.

Step 1. squared-distance potential

$$\phi(q) = \frac{1}{2} \hat{d}(q, q_{\text{rand}})^2$$

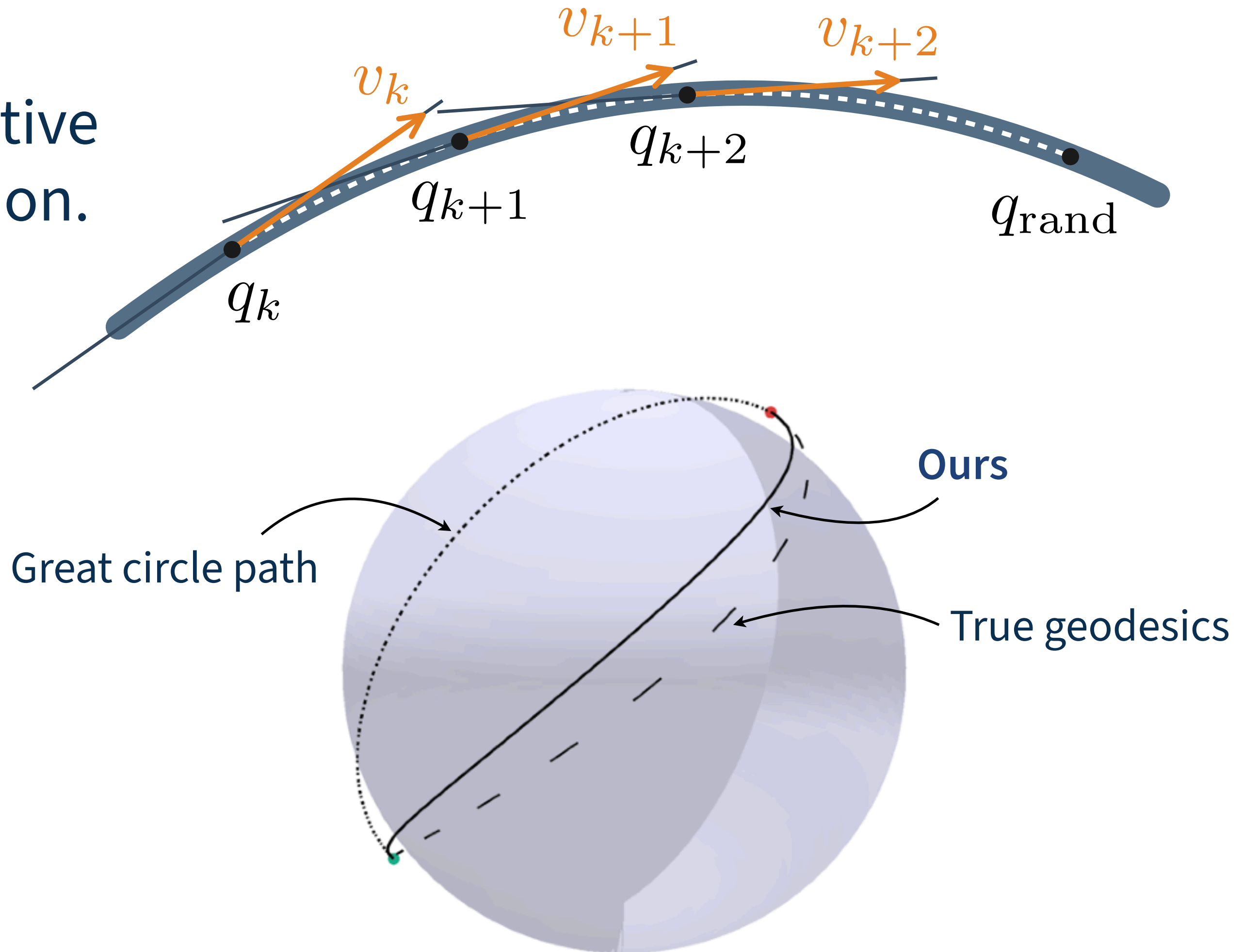
Step 2. Riemannian natural gradient

$$v_k = -G(q_k)^{-1} \nabla \phi(q_k)$$

Step 3. retraction step

$$\hat{v}_k = v_k / \|v_k\|_{q_k}$$

$$q_{k+1} = \mathcal{R}_{q_k}(s_k \hat{v}_k)$$



local planning on sphere under an anisotropic Riemannian metric